

B. String Mood

time limit per test: 0.25 seconds
memory limit per test: 256 megabytes

Limak can be either happy or sad. His mood changes (or stays same) when he reads an English uppercase letter. Letters **S** and **D** always make him sad, **H** makes him happy, and every vowel (**A E I O U**) flips his mood. Other letters do nothing.

Limak is happy right now. Among all 26^n possible strings with n uppercase English letters, count such strings that Limak will be happy after reading that string. Print the answer modulo $10^9 + 7$.

Input
An integer n ($1 \leq n \leq 10^{18}$).

Output
Print the answer modulo 1000000007.

Examples

input	Copy
1	
output	Copy
19	

input	Copy
2	
output	Copy
403	

input	Copy
11	
output	Copy
145418665	

Note
There are 19 strings of length $n = 1$ that will leave Limak happy: B, C, F, G, H, J, K, L, M, N, P, Q, R, T, V, W, X, Y, Z. The string "O" shouldn't be counted because the mood is switched from happy to sad.

For $n = 2$, there are 403 good strings: AA, AE, AH, AI, AO, AU, BB, ..., RZ, SA, SE, SH, SI, SO, SU, TB, TC, TF, ..., YQ, YR, YT, YV, YW, YX, YY, YZ, ZB, ZC, ZF, ZG, ZH, ZJ, ZK, ZL, ZM, ZN, ZP, ZQ, ZR, ZT, ZV, ZW, ZX, ZY, ZZ. The string "AO" is counted because the mood flips twice. Similarly, the string "SO" is counted because it first makes Limak sad (letter S) and then switches his mood from sad to happy (letter O).

Matrix Exponentiation

Finished

Practice



→ Virtual participation

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Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
336174837	Aug/29/2025 21:05	Happy New Year!